

Computer Architecture Lab (Week 5)

TP01: Write an assembly language to calculate $(a + b)^2$ and store answer in location (2005)

Sample Problem:

- Store Let $a = 09$ in memory location $2000H$ and $b = 0A$ in memory location $2001H$
- Calculate $(a + b)^2 = (13)^2 = 0169$
- Store $69H$ to memory location $2005H$ and $01H$ in memory location $2006H$

```
LDA 2000
MOV B A
LDA 2001
ADD B
MOV B A
MVI A 00
MOV C B
LOOP: ADD B
JNC NO_CARRY
INR D
NO_CARRY: DCR C
JNZ LOOP
STA 2005
MOV A D
STA 2006
```

```
HLT
```

TP02: Write an assembly language to calculate whether $m \times 2^n$ and store answer in address 2000

Sample Problem:

- Store Let $n = 1E$ in memory location $2000H$ and $m = 0A$ in memory location $2001H$
- Calculate $m \times 2^n = 0A \times 2(1E) = 0258$
- Store $58H$ to memory location $2005H$ and $02H$ in memory location $2006H$

```
LDA 2000
MOV B A
LDA 2001
MOV C A
MOV A B
ADD A
MOV B A
MVI A 00
MVI D 00
LOOP: ADD B
JNC NO_CARRY
INR D
NO_CARRY: DCR C
JNZ LOOP
STA 2005
MOV A D
STA 2006

HLT
```

TP03: Write an assembly language to calculate whether $(a+b)^2$ or $(a-b)^2$

Sample Problem:

- Store Let $a = 04H$, $b = 07H$, c (status) = $02H$
- Calculate $(a - b) ^ 2 = (04-07)^2= 09H$
- Store $09H$ to memory location $2005H$

LXI H 2000
MOV A M
INX H
MOV B M
INX H
MOV C M

MOV A C
CPI 01
JZ ADD
CPI 02
JZ SUB
JMP END

ADD: MOV A B
LXI H 2000
ADD M
JMP SQUARE

SUB: LXI H 2000
MOV A M
INX H
SUB M

SQUARE: MOV C A
MOV B A
MVI A 00
MOV D A

MULT: ADD C
JNC NO_CARRY
INR D
NO_CARRY: DCR B
JNZ MULT

STA 2005

END: HLT

**TP04: Write an assembly language to calculate
 $(a+b)^2 - 2b$ where $(a+b)^2 > 2b$**

- Let a in location @2000H, b in location @2001H
- Result store in location 2005H


```
LXI H 2000
MOV B M
INX H
MOV C M
MOV A C
ADD A
MOV D A
MOV A B
ADD C
MOV B A
MVI E 00
MOV E A
MVI C 00
```

```
MULT: ADD B
JNC NO_CARRY
INR C
NO_CARRY: DCR E
JNZ MULT
```

```
CMP D
JNC STORE_A
MOV A D
```

```
STORE_A: SUB D
STA 2005
HLT
```

TP05: Write an assembly language to calculate $(a-b)^2 + 3c$

- Let a in location @2000H, b in location @2001H, and c in location @2002H
- Result store in location 2005H

LXI H 2000
MOV B M
INX H
MOV C M
INX H
MOV D M
MOV A B
SUB C
MOV B A
MOV E A
MOV A D
ADD D
ADD D
MOV D A
MVI A 00
MVI C 00

MULT: ADD B
JNC NO_CARRY
INR C
NO_CARRY: DCR E
JNZ MULT

ADD D

STA 2005

HLT

TP06: Write an assembly language to calculate $(a-b)^2 + 4c$

- Let a in location @2000H, b in location @2001H, and c in location @2002H
- Result store in location 2005H

LXI H 2000
MOV B M
INX H
MOV C M
INX H
MOV D M
MOV A B
SUB C
MOV B A
MOV E A
MOV A D
ADD D
ADD D
ADD D
MOV D A
MVI A 00
MVI C 00

MULT: ADD B
JNC NO_CARRY
INR C
NO_CARRY: DCR E
JNZ MULT

ADD D

STA 2005

HLT

TP07: Write an assembly language to calculate the division of two 8-bit numbers. ($a > b$)

Sample Problem:

- Store Let $a = 1FH$, $b = 05H$
- Calculate $a/b = 06H$, and remain = $01H$
- Store $06H$ to memory location $2005H$ and remain $01H$ in memory location 2009

LXI H 2000
MOV A M
INX H
MOV B M
MVI C 00

DIVIDE: CMP B
JC END
SUB B
INR C
JMP DIVIDE

END: MOV D A
MOV A C
STA 2005
MOV A D
STA 2009
HLT

TP08: Write an assembly language to find minimum value among 5 numbers

Sample Problem:

- All 5 number store in memory location from 2000H to 2004H
- Length of series store in memory location 2005H
- Store the smallest number in memory location 2010H

LXI H 2005
MOV C M
DCR C
LXI H 2000
MOV A M
INX H

LOOP: CMP M
JC SKIP
MOV A M
SKIP: INX H
DCR C
JNZ LOOP

STA 2010
HLT

TP09: Write an assembly language to find maximum value among 7 numbers.

Sample Problem:

- All 7 number store in memory location from 2000H to 2006H
- Length of series store in memory location 2007H
- Store the biggest number in memory location 2010H

LXI H 2007
MOV C M
DCR C
LXI H 2000
MOV A M
INX H

LOOP: CMP M
JNC SKIP
MOV A M
SKIP: INX H
DCR C
JNZ LOOP

STA 2010
HLT

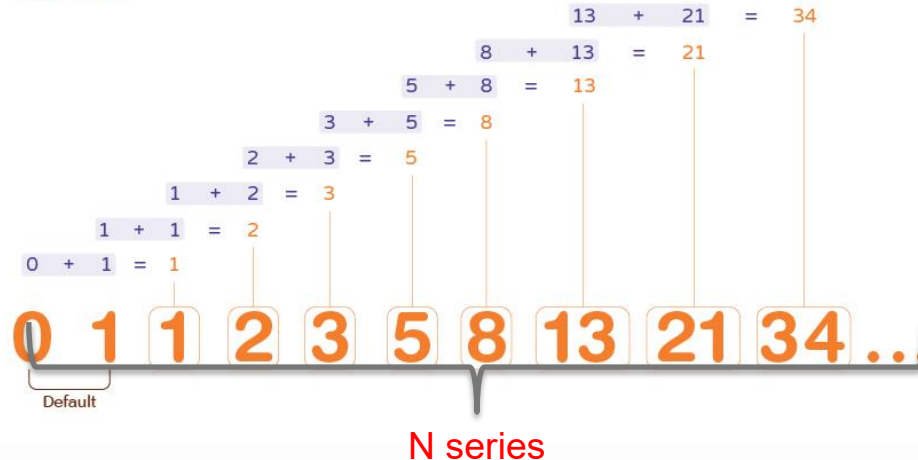
TP010: Write an assembly language to display Fibonacci numbers.

Store in any range of memory location. Ex N = 5 store from memory location @2000 to @2004.

Fibonacci Sequence

Numbers

MATH
MONKS



```
LXI H, 3000  
MVI C, 08  
MVI B, 00  
MVI D, 01  
MOV M, B  
INX H
```

```
FIB: MOV M, A  
MOV A, B  
ADD D  
MOV B, D  
MOV D, A  
INX H  
MOV M, A  
DCR C  
JNZ FIB
```

```
HLT
```

Thanks